Finite matrix composition equality forces orthogonality of profunctor creeds

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Editorial note. Official CC BY4.0 selected faithful editable transcription; original PDF/full source context and all37 numbered references retained.

Editorial note. Only Theorem20 counted; complete specialized supporting Lemma16 and Lemma17 proofs retained, no independent certification or new proof supplied.

Editorial note. Mathematical conditions and author finite-entry boundary retained; only standard TeX rendering and line layout supplied.

Editorial scope. Complete author material collection for one structural theorem; author retains responsibility for the mathematics.

Allowed standard prerequisites (editorial).

- Finite group coset/orbit cardinalities and coproduct distributivity of profunctors. Original source34:9–10 actual atom/coset construction and Propositions14/15 retained, citing standard treatment[17]. No recursive proof of ordinary group theory.
- Coend composition and nonnegative extended matrix sums. Original source34:4/34:6 definitions retained exactly as used in the author counting proof.

Author source: profunctors and matrix representation (34:4, 34:6–7)

We write I for the trivial category, which has one object $\star$ and one morphism $\mathrm{id}_\star$. For groups G and H, $G \leq H$ means that G is a subgroup of H. We write $[0,\infty]$ for the set $\{r \in \mathbb{R} \mid r \geq 0\} \cup \{\infty\}$. The sum $\sum_{j\in J} r_j$ of a possibly infinite family $(r_j)_{j\in J}$ of elements in $[0,\infty]$ is well-defined as the least upper bound of $\{\sum_{k\in K} r_k \mid K \subseteq J,\ K : \text{finite}\}$ (where $\infty + r = \infty$). A setoid is a set equipped with an equivalence relation. Given a setoid $(X, \approx_X)$, we write $[x]_\approx$ for the equivalence class $\{x' \in X \mid x \approx_X x'\}$. A groupoid A is a category in which every morphism is invertible. Then $A(a,a)$ is a group for every object $a \in A$. We use the diagrammatic order for composition: $(\alpha;\beta) \in A(a,c)$ when $\alpha \in A(a,b)$ and $\beta \in A(b,c)$. For groupoids A and B, a profunctor $P : A \nrightarrow B$ is a functor $P : A^{\mathrm{op}} \times B \to \mathsf{Set}$. For $x \in P(a,b)$, $\alpha \in A(a',a)$ and $\beta \in B(b,b')$, we

write $\alpha \cdot x \cdot \beta$ for $P(\alpha, \beta)(x)$. When $\beta = \mathrm{id}$, we simply write $\alpha \cdot x \cdot \mathrm{id}$ as $\alpha \cdot x$, and similarly for $x \cdot \beta := \mathrm{id} \cdot x \cdot \beta$. Profunctors $P : A \nrightarrow B$ and $Q : B \nrightarrow C$ compose analogously to relations:

$$(P;Q)(a,c) := \int^{b \in B} P(a,b) \times Q(b,c) := \left(\coprod_{b \in B} P(a,b) \times Q(b,c) \right) / \sim,$$

where $\sim$ is the equivalence relation defined by

$$\mathrm{in}_b(x,y) \sim \mathrm{in}_{b'}(x',y') \iff (x,y) = (x' \cdot \beta, \beta^{-1} \cdot y') \text{ for some } \beta \in B(b',b).$$

We shall simply write $\mathrm{in}_b(x,y)$ as (x,y). Then $(x \cdot \beta, y) \sim (x, \beta \cdot y)$ for every $x \in P(a,b)$, $\beta \in B(b,b')$ and $y \in Q(b',c)$. The groupoid action is defined by $\alpha \cdot [(x,y)]_\sim \cdot \beta := [(\alpha \cdot x, y \cdot \beta)]_\sim$. The weighted relational model with coefficient $[0,\infty]$ is written $\mathsf{WRel}_{[0,\infty]}$. Its objects are setoids and its morphisms from A to B are $((A/{\cong_A}) \times (B/{\cong_B}))$-matrices $R = (R_{[a],[b]})$ with $R_{[a],[b]} \in [0,\infty]$. Given $R \in \mathsf{WRel}_{[0,\infty]}(A,B)$ and $S \in \mathsf{WRel}_{[0,\infty]}(B,C)$, their composite is

$$(R;S)_{[a],[c]} := \sum_{[b] \in (B/{\cong_B})} R_{[a],[b]} S_{[b],[c]} := \sup \left\{ \sum_{[b] \in J} R_{[a],[b]} S_{[b],[c]} \;\middle|\; J \subseteq_{\mathrm{fin}} (B/{\cong_B}) \right\}.$$

Here $J \subseteq_{\mathrm{fin}} (B/{\cong_B})$ means a finite subset; $0 \cdot \infty := 0$ and $r \cdot \infty := \infty$ when $r > 0$.

Definition 6 (Matrix representation). We say a groupoid A is locally finite if $\#A(a,a) < \infty$ for every a. A factor of A is an assignment $\Phi^\oplus : \mathrm{obj}(A) \to (0,\infty)$ such that $\Phi^\oplus(a) = \Phi^\oplus(a')$ when $a \cong_A a'$. We define $\Phi^\ominus(a) := \#A(a,a)/\Phi^\oplus(a)$, so $\Phi^\oplus(a)\Phi^\ominus(a) = \#A(a,a)$. The matrix representation $\|P\|$ of a profunctor $P : A \nrightarrow B$ between locally finite groupoids A and B with factors is a matrix $\|P\| \in \mathsf{WRel}_{[0,\infty]}(\|A\|, \|B\|)$ given by

$$\|P\|_{[a],[b]} := \frac{\#P(a,b)}{\Phi^\ominus(a)\Psi^\oplus(b)}, \tag{2}$$

where $\|A\|$ is the setoid of the objects of A.

Author source: creeds and atomic decomposition (34:9–10)

Definition 9 (Creed [32]). Let $P : I \nrightarrow A$. The creed of P, written $\mathcal{C}P$, is a family of subsets $(\mathcal{C}P)_a \subseteq A(a,a)$ parameterised by objects $a \in A$ defined by

$$(\mathcal{C}P)_a := \{\alpha \in A(a,a) \mid \exists x \in P(a).\ x \cdot \alpha = x\} = \bigcup_{x \in P(a)} \mathrm{fix}(x),$$

where $\mathrm{fix}(x) \leq A^{\mathrm{op}}(a,a) \times B(b,b)$ for a profunctor $P : A \nrightarrow B$ and $x \in P(a,b)$ is the stabiliser subgroup, i.e.,

$$\mathrm{fix}(x) := \{(\alpha,\beta) \mid \alpha \in A(a,a),\ \beta \in B(b,b),\ \alpha \cdot x \cdot \beta = x\}.$$

The creed of $Q : A \nrightarrow I$ is defined similarly. A family $\mathcal{C} = (\mathcal{C}_a)_a$ of subsets $\mathcal{C}_a \subseteq A(a,a)$ is a creed if $\mathcal{C} = \mathcal{C}P$ for some $P : I \nrightarrow A$. Note that we do not define $\mathcal{C}P$ for general profunctors $P : A \nrightarrow B$ (with $A \neq I \neq B$).

Definition 12. Let $\mathcal{C}$ and $\mathcal{C}'$ be creeds on A. They are orthogonal, written $\mathcal{C} \perp \mathcal{C}'$, if $\mathcal{C}_a \cap \mathcal{C}'_a \subseteq \{\mathrm{id}\}$ for every a. Each hom-category $\mathsf{Prof}(A, B)$ has sums defined as follows. For profunctors $P_i : A \nrightarrow B$, the profunctor $\coprod_{i\in I} P_i : A \nrightarrow B$ is defined by $(\coprod_{i\in I} P_i)(a, b) := \coprod_{i\in I}(P_i(a, b))$ on objects and $(\coprod_{i\in I} P_i)(\alpha, \beta)(\mathrm{inj}(x)) := \mathrm{inj}(P_j(\alpha, \beta)(x))$ on morphisms, where $\mathrm{inj} : P_j(a, b) \to (\coprod_{i\in I} P(a, b))$ is the injection. A subgroup $G \leq A(a, a)^{\mathrm{op}} \times B(b, b)$ induces a profunctor $\widetilde{G} : A \nrightarrow B$ given by

$$\widetilde{G}(a', b') := \{\alpha \cdot G \cdot \beta \mid \alpha \in A(a', a),\ \beta \in B(b, b')\},$$
$$\widetilde{G}(\alpha, \beta) := X \mapsto (\alpha \cdot X \cdot \beta),$$

where for $X \subseteq A(a', a) \times B(b, b')$, $\alpha \in A(a'', a')$ and $\beta \in B(b', b'')$, the subset $(\alpha \cdot X \cdot \beta) \subseteq A(a'', a) \times B(b, b'')$ is defined by

$$(\alpha \cdot X \cdot \beta) := \{(\alpha; \alpha_0, \beta_0; \beta) \mid (\alpha_0, \beta_0) \in X\}.$$

Note that $\widetilde{G}$ has a single orbit, i.e., for every $x \in \widetilde{G}(a, b)$ and $y \in \widetilde{G}(a', b')$, there exist $\alpha \in A(a', a)$ and $\beta \in B(b, b')$ such that $y = \alpha \cdot x \cdot \beta$. This profunctor cannot be decomposed by $+$ any further. Every profunctor can be written as a sum of profunctors of the form $\widetilde{G}$. Given a profunctor $P : A \nrightarrow B$, let $(x_i \in P(a_i, b_i))_{i\in I}$ be a family of representatives of the equivalence classes defined by the $(A^{\mathrm{op}} \times B)$-action. That is, for every $x \in P(a, b)$, there exists a unique $i \in I$ such that $x = \alpha \cdot x_i \cdot \beta$ for some $\alpha \in A(a, a_i)$ and $\beta \in B(b_i, b)$. Then $P \cong \coprod_{i\in I} \widetilde{\mathrm{fix}(x_i)}$.

Proposition 14. A profunctor $P : A \nrightarrow B$ can be expressed as

$$P \cong \coprod_{i\in I} \widetilde{G}_i, \qquad \forall i \in I.\ G_i \leq A^{\mathrm{op}}(a_i, a_i) \times B(b_i, b_i)$$

for some family $(G_i)_{i\in I}$. Furthermore, $(\coprod_i \widetilde{G}_i); (\coprod_j \widetilde{H}_j) \cong \coprod_{i,j} (\widetilde{G}_i; \widetilde{H}_j)$.

Proposition 15. For $G \leq A^{\mathrm{op}}(a, a) \times B(b, b)$ and $H \leq B^{\mathrm{op}}(b', b') \times C(c, c)$, we have $(\widetilde{G}; \widetilde{H}) \neq \varnothing$ if and only if $b \cong b'$. Here $\varnothing : A \nrightarrow C$ is the profunctor such that $\varnothing(a, c) = \varnothing$. Note that the composite $(\widetilde{G}; \widetilde{H})$ for $G \leq A^{\mathrm{op}}(a, a) \times B(b, b)$ and $H \leq B^{\mathrm{op}}(b, b) \times C(c, c)$ is not necessarily of a single orbit. It can be a sum $\coprod_{i\in I} \widetilde{K}_i$ with $\#I > 1$.

Author source: complete single-orbit comparison (34:10–11)

Lemma 16. Let A, B, C be locally finite groupoids equipped with factors Φ, Ψ, Θ, and $K \leq A^{\mathrm{op}}(a, a) \times B(b, b)$ and $H \leq B^{\mathrm{op}}(b, b) \times C(c, c)$. Then, for every $a' \in A$ and $c' \in C$,

$$(\|\widetilde{K}\|; \|\widetilde{H}\|)_{[a'],[c']} \leq \|\widetilde{K}; \widetilde{H}\|_{[a'],[c']}.$$

The equality holds if and only if

$$\{\beta_1; \beta; \beta_1^{-1} \mid (\mathrm{id}, \beta) \in K\} \cap \{\beta_2; \beta; \beta_2^{-1} \mid (\beta, \mathrm{id}) \in H\} = \{\mathrm{id}\}$$

for every $\beta_1, \beta_2 \in B(b, b)$.

Proof. Assume $a' \in A$ and $c' \in C$. When $\neg(a' \cong a)$, then $\widetilde{K}(a', b') = \varnothing$ for every $b' \in B$. So we have $\|\widetilde{K}\|_{[a'],[b']} = 0$ for every b' and $\coprod_{b' \in B} \widetilde{K}(a', b') \times \widetilde{H}(b', c') = \varnothing$. The former implies $(\|\widetilde{K}\|; \|\widetilde{H}\|)_{[a'],[c']} = 0$, and the latter implies $\|\widetilde{K}; \widetilde{H}\|_{[a'],[c']} = 0$. So $\|\widetilde{K}\|; \|\widetilde{H}\|$ and $\|\widetilde{K}; \widetilde{H}\|$ coincide on $([a'], [c'])$-entry when $\neg(a' \cong a)$. The case that $\neg(c' \cong c)$ is similar. We calculate $(\|\widetilde{K}\|; \|\widetilde{H}\|)_{[a],[c]}$. Since $\widetilde{K}$ is the collection of cosets of K,

$$\#\widetilde{K}(a, b) = \frac{\#(A^{\mathrm{op}}(a, a) \times B(b, b))}{\#K}.$$

So

$$\|\widetilde{K}\|_{[a],[b]} = \frac{\#(A^{\mathrm{op}}(a, a) \times B(b, b))}{\#K\, \Phi^{\ominus}(a) \Psi^{\oplus}(b)} = \frac{\Phi^{\oplus}(a) \Psi^{\ominus}(b)}{\#K}.$$

The above formula, together with $\|\widetilde{K}\|_{[a],[b']} = 0$ for $\neg(b' \cong b)$, concludes

$$(\|\widetilde{K}\|; \|\widetilde{H}\|)_{[a],[c]} = \frac{\Phi^{\oplus}(a)\, \#B(b, b)\, \Theta^{\ominus}(c)}{\#K \#H}.$$

We calculate $\|\widetilde{K}; \widetilde{H}\|_{[a],[c]}$. Recall that $(\widetilde{K}; \widetilde{H})(a, c) \cong (\widetilde{K}(a, b) \times \widetilde{H}(b, c))/\!\sim$, where $(x, y) \sim (x', y')$ if and only if $(x \cdot \beta, \beta^{-1} \cdot y) = (x', y')$ for some $\beta \in B(b, b)$. For every $x \in \widetilde{K}(a, b)$ and $y \in \widetilde{H}(b, c)$, the equivalence class $[(x, y)]_{\sim} = \{(x \cdot \beta, \beta^{-1} \cdot y) \mid \beta \in B(b, b)\}$ has at most $\#B(b, b)$ elements. Therefore

$$\begin{aligned} \#(\widetilde{K}; \widetilde{H})(a, c) &\geq \frac{\#\widetilde{K}(a, b)\, \#\widetilde{H}(b, c)}{\#B(b, b)} \\ &= \frac{\#(A^{\mathrm{op}}(a, a) \times B(b, b))}{\#K} \times \frac{\#(B^{\mathrm{op}}(b, b) \times C(c, c))}{\#H} \times \frac{1}{\#B(b, b)}. \end{aligned}$$

So

$$\|\widetilde{K}; \widetilde{H}\|_{[a],[c]} = \frac{\#(\widetilde{K}; \widetilde{H})(a, c)}{\Phi^{\ominus}(a) \Theta^{\oplus}(c)} \geq \frac{\Phi^{\oplus}(a)\, \#B(b, b)\, \Theta^{\ominus}(c)}{\#K \#H}.$$

Assume the additional condition in the statement. Let $x = \alpha \cdot K \cdot \beta_1$ and $y = \beta_2 \cdot H \cdot \gamma$ and suppose $(x, y) = (x \cdot \beta, \beta^{-1} \cdot y)$. Then $\alpha \cdot K \cdot \beta_1 = \alpha \cdot K \cdot \beta_1 \cdot \beta$, so $K = K \cdot \beta_1 \cdot \beta \cdot \beta_1^{-1}$. Similarly, $\beta_2 \cdot H \cdot \gamma = \beta^{-1} \cdot \beta_2 \cdot H \cdot \gamma$ implies $H = \beta_2^{-1} \cdot \beta^{-1} \cdot \beta_2 \cdot H$. This means that $(\mathrm{id}, \beta_1; \beta; \beta_1^{-1}) \in K$ and $(\beta_2^{-1}; \beta^{-1}; \beta_2, \mathrm{id}) \in H$. By the additional assumption, $\beta = \mathrm{id}$. Therefore, $(x, y) = (x \cdot \beta, \beta^{-1} \cdot y)$ implies $\beta = \mathrm{id}$, and thus the equivalence class $[(x, y)]_{\sim}$ consists of $\#B(b, b)$ elements. So

$$\#(\widetilde{K}; \widetilde{H})(a, c) = \frac{\#\widetilde{K}(a, b)\, \#\widetilde{H}(b, c)}{\#B(b, b)}.$$

A straightforward calculation shows that $\|\widetilde{K}; \widetilde{H}\|_{[a],[c]} = (\|\widetilde{K}\|; \|\widetilde{H}\|)_{[a],[c]}$. Conversely, assume that $\{\beta_1; \beta; \beta_1^{-1} \mid (\mathrm{id}, \beta) \in K\} \cap \{\beta_2; \beta; \beta_2^{-1} \mid (\beta, \mathrm{id}) \in H\}$ contains $\beta \neq \mathrm{id}$ for some $\beta_1, \beta_2 \in B(b, b)$. Then $K = K \cdot \beta_1 \cdot \beta \cdot \beta_1^{-1}$ and hence $K \cdot \beta_1 = K \cdot \beta_1 \cdot \beta$. Similarly, since $H = \beta_2 \cdot \beta \cdot \beta_2^{-1} \cdot H$, we have $\beta^{-1} \cdot \beta_2^{-1} \cdot H = \beta_2^{-1} \cdot H$. Let $x = K \cdot \beta_1$ and $y = \beta_2^{-1} \cdot H$. Then $(x, y) = (x \cdot \beta, \beta^{-1} \cdot y)$, so the equivalence class $[(x, y)]_{\sim}$ consists of at most $\#B(b, b)/2$ elements. Recall the inequality

$$\#(\widetilde{K}; \widetilde{H})(a, c) \geq \frac{\#\widetilde{K}(a, b)\, \#\widetilde{H}(b, c)}{\#B(b, b)}.$$

Since $\#\widetilde{K}(a,b)$ and $\#\widetilde{H}(b,c)$ are finite, the equality holds only if all the equivalence classes $[(x,y)]_\sim$ consist of $\#B(b,b)$ elements. Therefore

$$\#(\widetilde{K};\widetilde{H})(a,c) > \frac{\#\widetilde{K}(a,b)\,\#\widetilde{H}(b,c)}{\#B(b,b)}$$

and hence $\|\widetilde{K};\widetilde{H}\|_{[a],[c]} > (\|\widetilde{K}\|;\|\widetilde{H}\|)_{[a],[c]}$. ◁

Author source: complete general-case comparison (34:11–12)

By Proposition 14, the argument for the single-orbit case extends to the general case.

Lemma 17. Let $P : A \nrightarrow B$ and $Q : B \nrightarrow C$, and assume factors for A, B, C. If $\mathcal{C}(P(a,-)) \perp \mathcal{C}(Q(-,c))$ for every $a \in A$ and $c \in C$, then $\|P\|;\|Q\| = \|P;Q\|$.

Proof. By Proposition 14, profunctors P and Q can be described as $P \cong \coprod_{i\in I} \widetilde{H}_i$ and $Q \cong \coprod_{j\in J} \widetilde{K}_j$ where $H_i \leq A^{\mathrm{op}}(a_i,a_i) \times B(b_i,b_i)$ and $K_j \leq B^{\mathrm{op}}(b_j,b_j)\times C(c_j,c_j)$. Note that $(P;Q) \cong \coprod_{i,j}(\widetilde{H}_i;\widetilde{K}_j)$ and $\|P;Q\| = \sum_{i,j}\|\widetilde{H}_i;\widetilde{K}_j\|$. So it suffices to show that $\|\widetilde{H}_i;\widetilde{K}_j\| = \|\widetilde{H}_i\|;\|\widetilde{K}_j\|$ for every i and j. We prove that the additional condition in Lemma 16 holds for every pair (H_i,K_j) such that $b_i \cong b_j$. We can assume without loss of generality that $b_i = b_j$. Let $b = b_i = b_j$ and $\beta_1,\beta_2 \in B(b,b)$. Assume that $\beta \in B(b,b)$ belongs to $\{\beta_1;\beta;\beta_1^{-1} \mid (\mathrm{id},\beta) \in H_i\} \cap \{\beta_2;\beta;\beta_2^{-1} \mid (\beta,\mathrm{id}) \in K_j\}$. Then $H_i \cdot \beta_1$ is a fixed-point of the action $(-)\cdot\beta$, so $\beta \in (\mathcal{C}(\widetilde{H}_i(a_i,-)))_b \subseteq (\mathcal{C}(P(a_i,-)))_b$. Similarly, since $\beta_2^{-1}\cdot K_j$ is a fixed-point of the action $\beta\cdot(-)$, we have $\beta \in (\mathcal{C}(\widetilde{K}_j(-,c_j)))_b \subseteq (\mathcal{C}(Q(-,c_j)))_b$. By the orthogonality assumption $\mathcal{C}(P(a_i,-)) \perp \mathcal{C}(Q(-,c_j))$, we have $\beta = \mathrm{id}$ as expected. ◁

Author source: completeness (34:12)

We have seen that the orthogonality of the symmetries of profunctors provides a sufficient condition for the matrix representation to behave functorially. In fact, this condition is almost necessary: as long as the result of matrix multiplication is finite, the functoriality of the matrix representation entails the orthogonality of the symmetry.

Theorem 20. Let $P : A \nrightarrow B$ and $Q : B \nrightarrow C$ be profunctors, and suppose that $\|P\|;\|Q\| = \|P;Q\|$. Then, for every $a \in A$ and $c \in C$, we have $\mathcal{C}(P(a,-)) \perp \mathcal{C}(Q(-,c))$ provided that $\|P;Q\|_{[a],[c]} < \infty$.

Proof. By Proposition 14, profunctors P and Q can be described as $P \cong \coprod_{i\in I} \widetilde{H}_i$ and $Q \cong \coprod_{j\in J} \widetilde{K}_j$ where $H_i \leq A^{\mathrm{op}}(a_i,a_i) \times B(b_i,b_i)$ and $K_j \leq B^{\mathrm{op}}(b_j,b_j)\times C(c_j,c_j)$. Note that $(P;Q) \cong \coprod_{i,j}(\widetilde{H}_i;\widetilde{K}_j)$ and $\|P;Q\| = \sum_{i,j}\|\widetilde{H}_i;\widetilde{K}_j\|$. Assume for contradiction that the creed orthogonality fails for some a and c.

Then the orthogonality fails for $(\widetilde{H}_{i_0}, \widetilde{K}_{j_0})$ for some (i_0, j_0). By Lemma 16, $(\|\widetilde{H}_{i_0}\|; \|\widetilde{K}_{j_0}\|)_{[a],[c]} < \|\widetilde{H}_{i_0}; \widetilde{K}_{j_0}\|_{[a],[c]}$, i.e.,

$$(\|\widetilde{H}_{i_0}\|; \|\widetilde{K}_{j_0}\|)_{[a],[c]} + \epsilon = \|\widetilde{H}_{i_0}; \widetilde{K}_{j_0}\|_{[a],[c]}$$

for some $\epsilon > 0$. Since $(\|\widetilde{H}_i\|; \|\widetilde{K}_j\|)_{[a],[c]} \leq \|\widetilde{H}_i; \widetilde{K}_j\|_{[a],[c]}$ for every i and j, we have

$$\left(\sum_{i,j} \|\widetilde{H}_i\|; \|\widetilde{K}_j\|\right)_{[a],[c]} + \epsilon \leq \left(\sum_{i,j} \|\widetilde{H}_i; \widetilde{K}_j\|\right)_{[a],[c]}.$$

Since $\sum_{i,j} \|\widetilde{H}_i; \widetilde{K}_j\| < \infty$, we have

$$\left(\sum_{i,j} \|\widetilde{H}_i\|; \|\widetilde{K}_j\|\right)_{[a],[c]} < \left(\sum_{i,j} \|\widetilde{H}_i; \widetilde{K}_j\|\right)_{[a],[c]},$$

a contradiction. ◁

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